Complexity Theory
Quiz 4 Set A

For each of the below statements answer whether it is True, False, or Unknown (if statement's truthness is not known). For each statement, you will get 1.5 **mark** for the right answer and $-(.5)$ for the wrong answer. Write the answers on this sheet and return it back. You do not need to submit the rough sheets. Make sure to read through all the questions, as they are **NOT** arranged in ascending order of difficulty.

Name: **Roll No:**

Duration: 40 Minutes **Marks:**

1. $\mathit{Palin} \in \mathsf{NC}^0$, where Palin is a set of palindromic binary strings.

False. Because any NC^0 circuit's output can depend only on constant number of input bits.

2. $\mathit{SAT} \in \mathsf{P}_{/\mathsf{poly}}$.

Unknown.

3. $\mathsf{NC}^1 \subseteq \mathsf{PSPACE}$.

False. NC^1 contains undecidable problem UHALT. (This question was tricky. :D)

4. $\mathit{SAT} \in \mathsf{BPP}$.

Unknown. If true, it will imply $\mathsf{NP} \subseteq \mathsf{BPP}$. If false, it will imply $\mathsf{NP} \not\subseteq \mathsf{BPP}$.

5. If L_1 and L_2 are in RP, then $L_1 \cap L_2$ is also in RP.

True. The RP machine for $L_1 \cap L_2$ will simulate RP machines of L_1 and L_2 and output 1 if and only if both the machines output 1. The probability analysis is left to you (it requires the fact that any constant more than 0 is good enough.)

6. $\mathsf{P} \subseteq \mathsf{ZPP}$.

True. Trivial.

7. The language L used in Kannan's theorem's proof is in $\mathsf{SIZE}(n^{c+1})$.

True. Follows from the way we chose circuit family C_n.

8. $\mathsf{ZPP} \not\subseteq \mathsf{BPP}$.

False. $\mathsf{ZPP} = \mathsf{RP} \cap \mathsf{coRP}$. And both RP and coRP are contained in BPP.

9. The final randomized algorithm (Algorithm D) of *ZEROP* that we saw in the class is an RP algorithm.
False. It is a coRP algorithm as it may make a mistake on “No" instances.

10. If $\mathsf{P} = \mathsf{NP}$, then $\mathsf{EXP} \not\subseteq \mathsf{P}_{/\mathsf{poly}}$.
True. $\mathsf{P} = \mathsf{NP}$ implies that $\Sigma_2^p = \mathsf{P}$ (hierarchy collapses to P). Now, $\mathsf{EXP} \not\subseteq \mathsf{P}_{/\mathsf{poly}}$ has to be true. Otherwise, if $\mathsf{EXP} \subseteq \mathsf{P}_{/\mathsf{poly}}$, then $\mathsf{EXP} = \Sigma_2^p$ (Meyer’s theorem). This will imply $\mathsf{P} = \mathsf{EXP}$, which is not true according to time hierarchy theorem.

Complexity Theory
Quiz 4 Set B

For each of the below statements answer whether it is True, False, or Unknown (if statement's truthness is not known). For each statement, you will get 1.5 **mark** for the right answer and $-(.5)$ for the wrong answer. Write the answers on this sheet and return it back. You do not need to submit the rough sheets. Make sure to read through all the questions, as they are **NOT** arranged in ascending order of difficulty.

Name: **Roll No:**
Duration: 40 Minutes **Marks:**

1. $\mathit{SAT} \in \mathsf{BPP}$.
Unknown.

2. $\mathsf{ZPP} \not\subseteq \mathsf{BPP}$.
False.

3. $\mathsf{NC}^1 \subseteq \mathsf{PSPACE}$.
False.

4. $\mathit{Palin} \in \mathsf{NC}^0$, where Palin is a set of palindromic binary strings.
False.

5. The final randomized algorithm (Algorithm D) of ZEROP that we saw in the class is an RP algorithm.
False.

6. $\mathsf{P} \subseteq \mathsf{ZPP}$.
True.

7. If $\mathsf{P} = \mathsf{NP}$, then $\mathsf{EXP} \not\subseteq \mathsf{P}_{/\mathsf{poly}}$.
True.

8. $\mathit{SAT} \in \mathsf{P}_{/\mathsf{poly}}$.
Unknown.

9. If L_1 and L_2 are in RP, then $L_1 \cap L_2$ is also in RP.
True.

10. The language L used in Kannan's theorem's proof is in $\mathsf{SIZE}(n^{c+1})$.
True.